

Table 1: NDCG performance of our approach compared to random ranking for Next Release Problem. The cell value $\alpha \pm \beta\%$ indicates that our approach achieves α on NDCG, which is $+\beta\%$ better or $-\beta\%$ worse than random ranking. Purple cells denote performance improvement, blue-gray cells denote performance degradation, while uncolored cells indicate no performance change compared to random ranking. $\dagger$ and $\star$ denote $p < 0.001$ and $0.001 \leq p < 0.05$, respectively.

Dataset	NDCG@1	NDCG@3	NDCG
nrp-e1	0.9571 ^{+55.8%*}	0.9917 ^{+50.6%*}	0.9928 ^{+15.5%*}
nrp-e2	1.0000 ^{+55.6%*}	0.9427 ^{+41.6%*}	0.9777 ^{+13.5%*}
nrp-e3	0.7286 ^{+21.4%}	0.8033 ^{+24.3%}	0.9216 ^{+8.1%}
nrp-e4	0.5714 ^{-21.6%*}	0.7366 ^{+4.4%}	0.8866 ^{+1.1%}
nrp-g1	0.9429 ^{+112.9%*}	0.9840 ^{+62.6%*}	0.9871 ^{+20.3%*}
nrp-g2	0.9857 ^{+64.3%*}	0.9859 ^{+46.7%*}	0.9883 ^{+14.9%*}
nrp-g3	0.9429 ^{+50.0%*}	0.9018 ^{+44.2%*}	0.9720 ^{+14.6%*}
nrp-g4	0.5000 ^{+2.9%}	0.5030 ^{-24.3%*}	0.8184 ^{-2.7%}
nrp-m1	0.9286 ^{+66.7%*}	0.9566 ^{+64.1%*}	0.9805 ^{+18.7%*}
nrp-m2	0.8857 ^{+55.0%*}	0.9416 ^{+50.4%*}	0.9713 ^{+15.8%*}
nrp-m3	0.9000 ^{+31.2%}	0.9056 ^{+29.8%*}	0.9660 ^{+11.0%*}
nrp-m4	0.9429 ^{+94.1%*}	0.9830 ^{+65.9%*}	0.9861 ^{+19.9%*}
nrp1	0.5714 ^{+14.3%}	0.6396 ^{+1.3%}	0.8378 ^{+0.5%}
nrp2	0.8571 ^{+39.5%*}	0.8085 ^{+26.9%*}	0.9344 ^{+10.2%*}
nrp3	0.8571 ^{+50.0%*}	0.7785 ^{+24.6%*}	0.9214 ^{+9.7%*}
nrp4	0.9000 ^{+37.0%*}	0.9141 ^{+49.9%*}	0.9661 ^{+14.5%*}
nrp5	0.4571 ^{-36.0%}	0.6928 ^{-2.7%}	0.8492 ^{-4.1%}
Average	0.8193 ^{+40.8%$\dagger$}	0.8511 ^{+33.0%$\dagger$}	0.9387 ^{+10.7%$\dagger$}

Table 2: NDCG performance of our approach compared to random ranking for Software Effort Estimation.

Dataset	NDCG@1	NDCG@3	NDCG
china-train_fold1	1.0000 ^{+48.9%*}	0.8772 ^{+21.2%*}	0.9725 ^{+10.6%*}
china-train_fold2	1.0000 ^{+45.8%*}	0.9771 ^{+36.0%*}	0.9854 ^{+11.8%*}
china-train_fold3	0.9571 ^{+116.1%*}	0.8967 ^{+65.8%*}	0.9655 ^{+19.5%*}
desharnais-train_fold1	0.4857 ^{-24.4%}	0.5728 ^{-2.5%}	0.8114 ^{-3.6%}
desharnais-train_fold2	0.9143 ^{+77.8%*}	0.8026 ^{+36.3%*}	0.9259 ^{+12.8%*}
desharnais-train_fold3	0.7143 ^{+22.0%}	0.7197 ^{+35.4%*}	0.8707 ^{+6.6%}
finnish-train_fold1	0.7429 ^{+44.4%*}	0.8065 ^{+35.9%*}	0.9206 ^{+11.1%*}
finnish-train_fold2	0.6714 ^{+34.3%}	0.6365 ^{-2.4%}	0.8658 ^{+2.9%}
finnish-train_fold3	0.5714 ^{-4.8%}	0.6891 ^{-1.2%}	0.8724 ^{-0.2%}
maxwell-train_fold1	0.8571 ^{+93.5%*}	0.9111 ^{+61.0%*}	0.9556 ^{+18.4%*}
maxwell-train_fold2	0.7714 ^{+22.7%}	0.7794 ^{+10.5%}	0.9244 ^{+6.9%*}
maxwell-train_fold3	0.7429 ^{+30.0%}	0.8927 ^{+41.4%*}	0.9391 ^{+11.6%*}
miyazaki-train_fold1	0.6571 ^{+2.2%}	0.6737 ^{+1.7%}	0.8780 ^{+1.7%}
miyazaki-train_fold2	0.7143 ^{+61.3%*}	0.8564 ^{+40.6%*}	0.9381 ^{+13.9%*}
miyazaki-train_fold3	1.0000 ^{+66.7%*}	0.8772 ^{+34.0%*}	0.9705 ^{+13.6%*}
Average	0.7867 ^{+42.4%*}	0.7979 ^{+27.6%*}	0.9197 ^{+9.2%$\dagger$}

Table 3: NDCG performance of our approach compared to random ranking for Software Configuration Tuning.

Dataset	NDCG@1	NDCG@3	NDCG
dnn_adiac	1.0000 ^{+46.3%*}	0.7822 ^{+16.0%*}	0.9272 ^{+6.6%*}
dnn_adiac_reverse	0.7714 ^{+38.5%}	0.6527 ^{+7.5%}	0.8888 ^{+6.8%}
dnn_coffee	1.0000 ^{+105.9%*}	0.7282 ^{+23.1%}	0.9181 ^{+12.3%*}
dnn_coffee_reverse	0.9857 ^{+64.3%*}	0.9913 ^{+48.2%*}	0.9931 ^{+16.1%*}
dnn_dsr	0.4000 ^{-22.6%}	0.4530 ^{-25.8%*}	0.7677 ^{-7.9%*}
dnn_dsr_reverse	0.7000 ^{+53.1%}	0.7714 ^{+41.5%*}	0.9178 ^{+13.8%*}
dnn_sa	0.5000 ^{-14.3%}	0.6168 ^{-5.7%}	0.8108 ^{-4.9%}
dnn_sa_reverse	0.9000 ^{+186.4%*}	0.9427 ^{+101.8%*}	0.9715 ^{+25.8%*}
llvm	0.8571 ^{+53.8%*}	0.7792 ^{+31.4%*}	0.9095 ^{+10.3%*}
llvm_reverse	0.7857 ^{+77.4%*}	0.9386 ^{+60.9%*}	0.9434 ^{+15.8%*}
lrzip	0.1429 ^{-73.7%*}	0.4851 ^{-20.7%*}	0.7512 ^{-10.1%*}
lrzip_reverse	0.8286 ^{+48.7%*}	0.9558 ^{+44.6%*}	0.9641 ^{+13.1%*}
mariadb	0.2000 ^{-53.3%}	0.5516 ^{-7.3%}	0.7686 ^{-5.8%}
mariadb_reverse	0.8571 ^{+66.7%*}	0.6886 ^{+5.3%}	0.8862 ^{+5.9%*}
mongodb	0.7143 ^{+25.0%}	0.8020 ^{+23.9%}	0.8923 ^{+5.8%}
mongodb_reverse	0.8286 ^{+23.4%}	0.7964 ^{+12.1%}	0.9264 ^{+5.7%*}
storm_rs	0.8571 ^{+100.0%*}	0.7471 ^{+19.2%}	0.9096 ^{+10.2%*}
storm_rs_reverse	0.5571 ^{+11.4%}	0.6820 ^{+19.3%}	0.8556 ^{+4.7%}
storm_wc	0.6000 ^{+0.0%}	0.5778 ^{-7.8%}	0.8423 ^{-0.5%}
storm_wc_reverse	0.7143 ^{+51.5%*}	0.7473 ^{+20.7%*}	0.8826 ^{+6.3%*}
trimesh	0.7143 ^{-15.3%}	0.7244 ^{+1.7%}	0.8805 ^{-1.8%}
trimesh_reverse	0.5429 ^{-17.4%}	0.6511 ^{-1.9%}	0.8369 ^{-2.0%}
vp9	0.9286 ^{+30.0%}	0.9409 ^{+27.5%*}	0.9783 ^{+9.1%*}
vp9_reverse	0.9143 ^{+30.6%}	0.8418 ^{+20.7%}	0.9515 ^{+8.6%*}
x264	0.7143 ^{+19.0%}	0.8683 ^{+33.4%*}	0.9338 ^{+10.0%*}
x264_reverse	0.8500 ^{+59.4%*}	0.9172 ^{+32.7%*}	0.9637 ^{+12.8%*}
Average	0.7255 ^{+34.4%*}	0.7551 ^{+20.1%†}	0.8951 ^{+6.4%†}

Table 4: NDCG performance of our approach compared to random ranking for Software Defect Prediction.

Dataset	NDCG@1	NDCG@3	NDCG
ant-1.7_J48	0.8857 ^{+87.9%*}	0.9650 ^{+68.8%*}	0.9609 ^{+18.3%*}
ant-1.7_KNN	0.3429 ^{-25.0%}	0.5598 ^{-3.4%}	0.7975 ^{-2.3%}
ant-1.7_LR	0.8571 ^{+36.4%*}	0.7719 ^{+14.4%}	0.9212 ^{+6.6%*}
ant-1.7_NB	0.7143 ^{+16.3%}	0.7645 ^{+32.5%*}	0.8921 ^{+7.0%}
camel-1.6_J48	0.6571 ^{+15.0%}	0.8315 ^{+29.0%*}	0.9021 ^{+6.1%*}
camel-1.6_KNN	0.5571 ^{-11.4%}	0.6067 ^{-6.2%}	0.8406 ^{-1.4%}
camel-1.6_LR	0.7143 ^{+13.6%}	0.6644 ^{+6.2%}	0.8620 ^{+1.6%}
camel-1.6_NB	0.8143 ^{+32.6%}	0.8162 ^{+18.9%}	0.9118 ^{+7.0%}
ivy-2.0_J48	0.4143 ^{-32.6%}	0.5273 ^{-19.7%}	0.7861 ^{-7.8%}
ivy-2.0_KNN	0.4143 ^{-14.7%}	0.5513 ^{-11.2%}	0.8079 ^{-2.1%}
ivy-2.0_LR	0.5571 ^{-9.3%}	0.5349 ^{-19.8%}	0.8213 ^{-4.3%}
ivy-2.0_NB	1.0000 ^{+100.0%*}	0.9088 ^{+38.5%*}	0.9760 ^{+15.1%*}
jedit-4.0_J48	0.4000 ^{-33.3%}	0.4658 ^{-26.3%}	0.7706 ^{-8.6%*}
jedit-4.0_KNN	0.2857 ^{-55.6%*}	0.4548 ^{-31.5%*}	0.7665 ^{-10.3%*}
jedit-4.0_LR	0.2857 ^{-44.4%*}	0.5454 ^{-5.1%}	0.7903 ^{-3.8%}
jedit-4.0_NB	0.7143 ^{-3.8%}	0.6383 ^{-5.8%}	0.8629 ^{-1.4%}
lucene-2.4_J48	0.7143 ^{+0.0%}	0.6971 ^{+0.2%}	0.8834 ^{+0.9%}
lucene-2.4_KNN	0.6000 ^{+20.0%}	0.7295 ^{+16.9%}	0.8664 ^{+4.2%}
lucene-2.4_LR	0.8571 ^{+81.8%*}	0.6191 ^{+1.5%}	0.8661 ^{+4.5%}
lucene-2.4_NB	0.7143 ^{+6.4%}	0.6723 ^{-1.0%}	0.8894 ^{+3.0%}
poi-3.0_J48	0.7286 ^{+18.6%}	0.9305 ^{+49.5%*}	0.9292 ^{+10.0%*}
poi-3.0_KNN	0.4143 ^{-43.1%*}	0.5478 ^{-22.6%*}	0.8038 ^{-8.1%*}
poi-3.0_LR	0.2857 ^{-37.5%}	0.5952 ^{-3.3%}	0.7989 ^{-3.5%}
poi-3.0_NB	0.1429 ^{-63.0%*}	0.5131 ^{-10.1%}	0.7678 ^{-4.7%*}
synapse-1.2_J48	0.2857 ^{-54.5%*}	0.6514 ^{+8.6%}	0.8143 ^{-3.3%}
synapse-1.2_KNN	1.0000 ^{+52.2%*}	0.7878 ^{+17.1%*}	0.9411 ^{+9.3%*}
synapse-1.2_LR	0.4286 ^{-14.3%}	0.6891 ^{+23.2%}	0.8405 ^{+3.5%}
synapse-1.2_NB	0.1429 ^{-71.4%*}	0.5484 ^{-14.0%}	0.7766 ^{-6.8%}
velocity-1.6_J48	0.9429 ^{+43.5%}	0.8398 ^{+20.1%}	0.9506 ^{+8.7%}
velocity-1.6_KNN	0.5429 ^{-22.4%}	0.8015 ^{+9.9%}	0.8851 ^{+0.2%}
velocity-1.6_LR	1.0000 ^{+94.4%*}	1.0000 ^{+48.8%*}	0.9933 ^{+17.6%*}
velocity-1.6_NB	0.4000 ^{-40.4%}	0.6439 ^{-6.6%}	0.8315 ^{-3.8%}
xalan-2.6_J48	0.9286 ^{+58.5%*}	0.8021 ^{+27.4%*}	0.9373 ^{+10.5%*}
xalan-2.6_KNN	0.6857 ^{+45.5%*}	0.6864 ^{+12.2%}	0.8849 ^{+7.2%*}
xalan-2.6_LR	0.8714 ^{+60.5%*}	0.8621 ^{+36.5%*}	0.9516 ^{+13.4%*}
xalan-2.6_NB	0.5714 ^{+60.0%*}	0.5466 ^{+1.0%}	0.8138 ^{+3.4%}
xerces-1.4_J48	0.4286 ^{-23.1%}	0.5514 ^{-1.1%}	0.8191 ^{-1.1%}
xerces-1.4_KNN	0.5714 ^{-4.8%}	0.6252 ^{-8.1%}	0.8368 ^{-3.0%}
xerces-1.4_LR	0.9143 ^{+42.2%*}	0.8704 ^{+24.0%*}	0.9614 ^{+11.0%*}
xerces-1.4_NB	0.8571 ^{+46.3%*}	0.9609 ^{+56.5%*}	0.9742 ^{+15.7%*}
Average	0.6161 ^{+8.2%}	0.6944 ^{+9.1%*}	0.8672 ^{+2.7%}

Table 5: NDCG performance of our approach compared to random ranking for Software Product Lines Testing.

Dataset	NDCG@1	NDCG@3	NDCG
7z	0.8857 ^{+72.2%*}	0.9244 ^{+48.6%*}	0.9584 ^{+15.0%*}
Amazon	1.0000 ^{+141.4%*}	0.7548 ^{+38.8%*}	0.9312 ^{+15.8%*}
BerkeleyDBC	0.4714 ^{-26.7%}	0.7637 ^{+8.0%}	0.8647 ^{-0.4%}
CocheEcologico	0.8571 ^{+53.8%*}	0.7951 ^{+19.0%*}	0.9317 ^{+9.7%*}
CounterStrikeSimpleFeatureModel	0.5429 ^{+11.8%}	0.6984 ^{+14.9%}	0.8630 ^{+4.9%}
DSSample	0.8857 ^{+34.8%*}	0.9332 ^{+37.0%*}	0.9574 ^{+10.1%*}
Drupal	0.5571 ^{+0.0%}	0.6820 ^{+4.0%}	0.8668 ^{+2.2%}
Dune	0.1429 ^{-76.7%*}	0.3771 ^{-42.5%*}	0.7315 ^{-15.7%*}
ElectronicDrum	1.0000 ^{+100.0%*}	0.8248 ^{+32.4%*}	0.9598 ^{+14.9%*}
HiPAcc	0.8857 ^{+100.0%*}	0.9689 ^{+66.2%*}	0.9756 ^{+20.4%*}
JHipster	0.8429 ^{+40.5%*}	0.7409 ^{+10.6%}	0.9129 ^{+6.6%*}
JavaGC	0.8000 ^{+14.3%}	0.6979 ^{+7.0%}	0.9021 ^{+5.4%}
ModelTransformation	0.7571 ^{+35.9%}	0.7301 ^{+15.7%}	0.9087 ^{+8.1%*}
Polly	0.6857 ^{+45.5%*}	0.7651 ^{+23.0%*}	0.9036 ^{+8.9%*}
SPLSSimuelESPnP	0.6143 ^{+0.0%}	0.7331 ^{+20.9%*}	0.8860 ^{+5.0%}
SmartHomev2.2	0.6571 ^{+53.3%}	0.8315 ^{+50.0%*}	0.9164 ^{+13.9%*}
VP9	0.6000 ^{+23.5%}	0.6794 ^{+11.6%}	0.8707 ^{+5.8%}
VideoPlayer	0.9429 ^{+65.0%*}	0.8252 ^{+22.1%}	0.9551 ^{+11.3%*}
WebPortal	0.9714 ^{+47.8%*}	0.9307 ^{+31.4%*}	0.9765 ^{+12.3%*}
lrzip	0.4714 ^{-26.7%}	0.7825 ^{+21.0%}	0.8605 ^{+1.1%}
x264	0.8571 ^{+57.9%*}	0.6253 ^{+12.4%}	0.8602 ^{+5.7%*}
Average	0.7347 ^{+36.6%*}	0.7650 ^{+21.5%†}	0.9044 ^{+7.7%†}

Table 6: NDCG performance of our approach compared to random ranking for the Software Project Scheduling. The cell value $\alpha^{\pm\beta\%}$ indicates that our approach achieves α on NDCG, is $+\beta\%$ better or $-\beta\%$ worse than random ranking. **Purple cells** denote performance improvement, **blue-gray cells** denote performance degradation, **uncolored cells** indicate no performance change compared to random ranking.[†] and * denote $p < 0.001$ and $0.001 \leq p < 0.05$, respectively.

Dataset	NDCG@1	NDCG@3	NDCG
10-10-skill-4-5	0.8571 ^{+50.0%*}	0.7844 ^{+22.8%*}	0.9233 ^{+9.1%*}
10-10-skill-6-7	0.7857 ^{+41.0%}	0.7915 ^{+28.9%*}	0.9240 ^{+10.2%*}
10-15-skill-4-5	0.7000 ^{+14.0%}	0.8144 ^{+24.9%}	0.9243 ^{+8.3%*}
10-15-skill-6-7	0.8571 ^{+11.1%}	0.8133 ^{+14.7%*}	0.9263 ^{+4.7%}
10-5-skill-4-5	0.5714 ^{-4.8%}	0.8534 ^{+28.5%*}	0.9014 ^{+5.3%}
10-5-skill-6-7	0.3286 ^{-43.9%*}	0.5311 ^{-20.7%*}	0.7924 ^{-7.2%*}
20-10-skill-4-5	0.5714 ^{+29.0%}	0.7616 ^{+33.4%*}	0.8906 ^{+9.7%*}
20-10-skill-6-7	0.8571 ^{+57.9%*}	0.8754 ^{+28.7%*}	0.9552 ^{+12.8%*}
20-15-skill-4-5	0.8857 ^{+59.0%*}	0.9214 ^{+58.1%*}	0.9644 ^{+16.7%*}
20-15-skill-6-7	0.8571 ^{+50.0%*}	0.9228 ^{+46.9%*}	0.9654 ^{+15.4%*}
20-5-skill-4-5	0.6000 ^{-4.5%}	0.7738 ^{+13.8%}	0.8860 ^{+3.1%}
20-5-skill-6-7	0.7143 ^{+47.1%*}	0.9346 ^{+51.2%*}	0.9420 ^{+13.1%*}
30-10-skill-4-5	0.8571 ^{+122.2%*}	0.8599 ^{+44.1%*}	0.9462 ^{+16.7%*}
30-10-skill-6-7	0.7000 ^{-2.0%}	0.8015 ^{+21.2%*}	0.9198 ^{+6.2%}
30-15-skill-4-5	0.8571 ^{+50.0%*}	0.8851 ^{+34.4%*}	0.9578 ^{+12.7%*}
30-15-skill-6-7	0.5857 ^{-8.9%}	0.7107 ^{+16.5%}	0.8832 ^{+4.9%}
30-5-skill-4-5	0.7143 ^{+22.0%}	0.9367 ^{+55.6%*}	0.9428 ^{+12.8%*}
30-5-skill-6-7	0.6571 ^{-2.1%}	0.7887 ^{+13.5%*}	0.9095 ^{+4.5%}
Average	0.7198 ^{+27.1%*}	0.8200 ^{+28.7%†}	0.9197 ^{+8.8%†}

Table 7: NDCG performance of our approach compared to random ranking for Third Party Library Migration.

Dataset	NDCG@1	NDCG@3	NDCG
migrationRule10_ALL	0.6714 ^{+11.9%}	0.8056 ^{+29.8%*}	0.9005 ^{+6.6%}
migrationRule10_CO	0.5714 ^{+33.3%}	0.8138 ^{+42.8%*}	0.9062 ^{+12.1%*}
migrationRule10_DS	0.3429 ^{-36.8%}	0.6819 ^{+8.8%}	0.8394 ^{-0.5%}
migrationRule10_MS	0.7286 ^{+41.7%*}	0.8365 ^{+33.9%*}	0.9185 ^{+10.2%*}
migrationRule18_ALL	0.8571 ^{+62.2%*}	0.9663 ^{+48.6%*}	0.9738 ^{+15.2%*}
migrationRule18_CO	0.5857 ^{+7.9%}	0.6038 ^{+16.4%}	0.8372 ^{+3.6%}
migrationRule18_DS	0.3667 ^{-36.8%}	0.6566 ^{+4.2%}	0.8118 ^{-2.3%}
migrationRule18_MS	0.5286 ^{+19.4%}	0.6946 ^{+21.9%}	0.8693 ^{+8.1%}
migrationRule1_ALL	0.6857 ^{+26.3%}	0.7162 ^{+14.2%}	0.8888 ^{+7.4%*}
migrationRule1_CO	0.8571 ^{+81.8%*}	0.7957 ^{+36.0%*}	0.9270 ^{+13.9%*}
migrationRule1_DS	0.7000 ^{-7.5%}	0.8467 ^{+12.5%*}	0.9256 ^{+3.5%}
migrationRule1_MS	0.6143 ^{+19.4%}	0.8280 ^{+39.5%*}	0.9173 ^{+11.1%*}
migrationRule2_ALL	0.6286 ^{+41.9%}	0.6196 ^{+3.7%}	0.8577 ^{+5.4%}
migrationRule2_CO	1.0000 ^{+118.8%*}	0.8351 ^{+38.2%*}	0.9585 ^{+16.0%*}
migrationRule2_DS	0.6571 ^{+24.3%}	0.6622 ^{+18.0%}	0.8813 ^{+7.3%*}
migrationRule2_MS	1.0000 ^{+59.1%*}	0.9139 ^{+40.2%*}	0.9808 ^{+15.0%*}
migrationRule3_ALL	0.5714 ^{+42.9%}	0.8059 ^{+53.6%*}	0.8944 ^{+13.1%*}
migrationRule3_CO	0.9286 ^{+103.1%*}	0.7991 ^{+43.5%*}	0.9277 ^{+14.2%*}
migrationRule3_MS	0.9429 ^{+34.7%*}	0.9513 ^{+27.3%*}	0.9772 ^{+10.6%*}
migrationRule4_ALL	0.4286 ^{-14.3%}	0.6221 ^{+1.9%}	0.8433 ^{+1.5%}
migrationRule4_CO	0.6429 ^{+25.0%}	0.8159 ^{+31.6%}	0.9003 ^{+8.3%}
migrationRule4_DS	0.6429 ^{-11.8%}	0.8200 ^{+17.8%*}	0.9060 ^{+3.1%}
migrationRule4_MS	0.7143 ^{+6.4%}	0.8269 ^{+16.9%*}	0.9167 ^{+3.8%}
migrationRule5_ALL	0.8571 ^{+39.5%*}	0.7543 ^{+24.2%*}	0.9228 ^{+10.0%*}
migrationRule5_CO	0.6714 ^{+17.5%}	0.8293 ^{+47.5%*}	0.9183 ^{+11.1%*}
migrationRule5_DS	0.5429 ^{+31.0%}	0.7000 ^{+24.4%*}	0.8633 ^{+6.6%*}
migrationRule5_MS	0.6143 ^{+10.3%}	0.8401 ^{+41.4%*}	0.9160 ^{+10.4%*}
migrationRule7_ALL	0.8571 ^{+87.5%*}	0.9372 ^{+60.2%*}	0.9715 ^{+19.6%*}
migrationRule7_CO	0.7429 ^{+62.5%*}	0.7669 ^{+38.0%*}	0.9068 ^{+12.0%*}
migrationRule7_DS	0.8714 ^{+117.9%*}	0.9134 ^{+74.1%*}	0.9659 ^{+21.7%*}
migrationRule7_MS	0.8571 ^{+57.9%*}	0.8000 ^{+31.5%*}	0.9394 ^{+12.6%*}
migrationRule8_ALL	0.7143 ^{+66.7%*}	0.8217 ^{+44.1%*}	0.9217 ^{+13.5%*}
migrationRule8_CO	0.7143 ^{+28.2%}	0.9346 ^{+54.8%*}	0.9470 ^{+13.5%*}
migrationRule8_DS	0.5714 ^{-27.3%*}	0.6507 ^{-11.6%}	0.8435 ^{-5.2%*}
migrationRule8_MS	0.7143 ^{+19.0%}	0.8217 ^{+22.9%*}	0.9308 ^{+9.0%*}
Average	0.6970 ^{+33.2%†}	0.7911 ^{+30.1%†}	0.9087 ^{+9.2%†}

Table 8: NDCG performance of our approach compared to random ranking for Workflow Scheduling.

Dataset	NDCG@1	NDCG@3	NDCG
CyberShake_100	0.7857 ^{+44.7%}	0.6433 ^{+8.9%}	0.8690 ^{+4.6%}
CyberShake_1000	0.8571 ^{+62.2%}	0.7664 ^{+24.5%*}	0.9166 ^{+10.4%*}
CyberShake_30	1.0000 ^{+100.0%*}	0.9090 ^{+54.6%*}	0.9754 ^{+18.6%*}
CyberShake_50	0.8143 ^{+62.9%*}	0.7184 ^{+21.2%*}	0.9123 ^{+10.5%*}
Epigenomics_100	1.0000 ^{+42.9%*}	0.8217 ^{+12.1%}	0.9449 ^{+7.4%*}
Epigenomics_24	0.7143 ^{+38.9%*}	0.8198 ^{+55.2%*}	0.9180 ^{+13.3%*}
Epigenomics_46	1.0000 ^{+45.8%*}	0.8772 ^{+28.3%*}	0.9699 ^{+11.4%*}
Epigenomics_997	0.5143 ^{+9.1%}	0.7493 ^{+19.6%}	0.8669 ^{+3.4%}
Inspiral_100	0.7143 ^{+22.0%}	0.7917 ^{+29.2%*}	0.9123 ^{+7.5%*}
Inspiral_1000	0.8571 ^{+93.5%*}	0.9315 ^{+61.9%*}	0.9646 ^{+19.3%*}
Inspiral_30	0.5714 ^{+5.3%}	0.6882 ^{+13.8%}	0.8584 ^{+3.3%}
Inspiral_50	0.8571 ^{+50.0%*}	0.6684 ^{-0.5%}	0.8836 ^{+3.4%}
Montage_100	0.5714 ^{-7.0%}	0.4797 ^{-25.5%*}	0.8028 ^{-5.4%}
Montage_1000	1.0000 ^{+29.6%}	0.9473 ^{+28.5%*}	0.9850 ^{+10.2%*}
Montage_25	0.6857 ^{+20.0%}	0.7014 ^{+7.0%}	0.8803 ^{+4.2%}
Montage_50	1.0000 ^{+37.3%*}	0.9951 ^{+45.6%*}	0.9872 ^{+12.8%*}
Sipht_100	0.8571 ^{+33.3%*}	0.8170 ^{+12.5%}	0.9283 ^{+5.8%*}
Sipht_1000	0.6857 ^{+23.1%}	0.8291 ^{+29.4%*}	0.9057 ^{+7.1%}
Sipht_30	0.4286 ^{-18.9%}	0.6267 ^{-1.3%}	0.8122 ^{-2.7%}
Sipht_60	0.2857 ^{-48.7%*}	0.6039 ^{-0.9%}	0.7977 ^{-4.8%}
Average	0.7600 ^{+32.3%†}	0.7692 ^{+21.2%†}	0.9046 ^{+7.0%†}

Table 9: NDCG performance of our approach compared to random ranking for Web Service Composition.

Dataset	NDCG@1	NDCG@3	NDCG
workflow_1	0.4714 ^{-15.4%}	0.6954 ^{+5.1%}	0.8492 ^{-0.1%}
workflow_1_reverse	0.6714 ^{+14.6%}	0.5759 ^{-17.5%*}	0.8257 ^{-3.8%}
workflow_10	0.9429 ^{+83.3%*}	0.9038 ^{+48.2%*}	0.9670 ^{+15.9%*}
workflow_10_reverse	0.7143 ^{+16.3%}	0.7901 ^{+13.3%}	0.9025 ^{+4.0%}
workflow_2	0.8571 ^{+71.4%*}	0.9572 ^{+58.2%*}	0.9682 ^{+16.4%*}
workflow_2_reverse	0.8143 ^{+72.7%*}	0.6591 ^{-4.1%}	0.8785 ^{+3.9%}
workflow_3	0.8571 ^{+15.4%}	0.8229 ^{+20.0%}	0.9248 ^{+5.3%}
workflow_3_reverse	0.4286 ^{-43.4%*}	0.7554 ^{+0.4%}	0.8590 ^{-4.3%}
workflow_4	0.9714 ^{+88.9%*}	0.9643 ^{+71.6%*}	0.9861 ^{+21.8%*}
workflow_4_reverse	0.8143 ^{+16.3%}	0.8456 ^{+18.4%*}	0.9264 ^{+5.3%}
workflow_5	0.5714 ^{-4.8%}	0.8583 ^{+37.8%*}	0.9030 ^{+6.8%*}
workflow_5_reverse	0.6143 ^{+34.4%}	0.8142 ^{+37.0%*}	0.9005 ^{+10.3%*}
workflow_6	0.8571 ^{+33.3%}	0.7129 ^{+14.2%*}	0.9094 ^{+7.8%*}
workflow_6_reverse	0.5429 ^{+46.2%}	0.7686 ^{+39.9%*}	0.8735 ^{+9.5%*}
workflow_7	0.4857 ^{-15.0%}	0.7237 ^{+16.4%}	0.8699 ^{+3.5%}
workflow_7_reverse	0.3714 ^{-43.5%}	0.4875 ^{-33.9%*}	0.7983 ^{-9.4%*}
workflow_8	0.6000 ^{+0.0%}	0.7452 ^{+11.4%}	0.8752 ^{+2.4%}
workflow_8_reverse	0.5714 ^{-7.0%}	0.7228 ^{+7.6%}	0.8619 ^{+0.8%}
workflow_9	0.4286 ^{-6.3%}	0.6495 ^{+4.0%}	0.8324 ^{+1.1%}
workflow_9_reverse	0.7000 ^{+14.0%}	0.8380 ^{+32.0%*}	0.9105 ^{+7.3%}
Average	0.6643 ^{+18.6%}	0.7645 ^{+19.0%*}	0.8911 ^{+5.2%*}